

Magnephy AB – Investment Memo

Magnephy is a bet that advanced optical microscopy, daily used in the academia context, becomes a quantitative tool for pharmaceutical R&D. Magnephy's technology relies on a unique tissue clearing technique combined with 3D Stimulated Emission Depletion (STED) and AI-based image segmentation in order to, for example, localise and quantify drug delivery or study tissue response at the subcellular level in intact tissue.

Core investment question: Can Magnephy convert a technically differentiated but specialized microscopy capability into a platform accessible for R&D with sufficient attraction to keep customers and defend itself against large microscopy vendors who are also entering the market?

1. Company Snapshot

Magnephy is a Swedish deep tech life science company founded as a spin-off from KTH/SciLife in 2023. It provides services to companies regarding tissue clearing, 3D Stimulated Emission Depletion (STED) imaging and AI-based quantitative analysis. Their expertise lies mainly in the analysis of kidney tissue, but they state the same technology can be applied to other tissue types.

2. Core Problem

The core problem lies in the current limitation faced by pharmaceutical companies when it comes to using microscopy to determine things like whether a candidate is working in the intended manner before continuing to put more money into it. One of the reasons behind this limitation is the conventional histology pipeline that relies on the physical sectioning of samples. This makes the processing practical and scalable, but spatial information can be lost.

The solution Magnephy offers is the 3D imaging of intact tissue that can recover spatial information. This, combined with the use of STED, provides nanoscale resolution. The company's published data demonstrates that their approach, applied to human kidney biopsies, can distinguish morphological differences between healthy and diseased tissue.

3. Technology & Scientific Differentiation

Magnephy's technological uniqueness relies on three components. The first is tissue clearing, developed in academic research, which enables them to run a rapid clearing and swelling protocol that allows tri-dimensional imaging of kidney tissue. The second is their access to STED microscopy, which provides super-resolution and allows structures below the conventional diffraction limit to be resolved. Finally, AI segmentation, which also comes from their academic research, has shown successful deep-learning segmentation of podocyte foot processes from human kidney biopsies.

4. Market & Competitive Landscape

4.1 Industry structure

R&D within the pharmaceutical industry is quickly adopting AI as part of their pipeline in drug design and testing. This is in conjunction with other technologies such as spatial biology and advanced microscopy that Magnephy focuses on. There are, nevertheless, several competitors that are using different approaches to offer a service that represents a relatively narrow segment.

4.2 Competitive axes

Several companies already perform 3D tissue imaging services and large microscopy companies such as ZEISS are increasingly introducing AI into their software and combining it with 3D imaging.

4.3 Competitor positioning

Company	Positioning	Advantage	Risk
ZEISS / Leica / Nikon	Microscopy hardware + software	Massive installed base, optical expertise, distribution and customer relationships	Could incorporate more advanced 3D/AI workflows into existing platforms
Internal pharma microscopy teams	In-house imaging and pathology	Control, speed and accumulated internal data	Customers may eventually internalize workflows if technology becomes standardized
Translucence Biosystems	Tissue clearing + light-sheet imaging + AI quantification	Broad workflow and modular services; established 3D tissue ecosystem	Similar outsourced-service model

4.4 Magnephy positioning

Strengths

- Strong academic and scientific credibility with founders experts in advanced microscopy, kidney biology and AI.
- Built on established research with over a decade of experience in the field
- Ability to preserve intact tissue architecture
- Potential applications beyond kidney disease
- Already significant success in pilot projects with industry clients (Q32 Bio)

Weaknesses

- Technology requires special instruments and expertise
- High competition surrounding AI & 3D imaging in microscopy
- Not every 3D imaging may be best approached with STED
- Adoption by pharmaceutical companies probably relies on reproducibility and assay validation which is not yet available.
- Limited public information regarding pricing, costs and whether companies are being kept as clients after first use

It is clear from its publications in Kidney International (2016, 2021 & 2023) that Magnephy has differentiating technology available and the expertise to carry out the whole imaging process successfully. The main question I see is whether there is a gain for pharmaceutical companies to switch towards this new technology. This depends on, for example, not only

showing a high-resolution image of a tissue but also showing how an antibody reaches a target of interest deep inside that tissue.

4.5 How the market could look in 5–7 years

Scenario A: Specialized Spatial Biology Platform

Magnephy establishes itself as a specialist provider of nanoscale 3D tissue assays for pharmaceutical development. Pharmaceutical companies use Magnephy's technology to measure things like tissue penetration or cellular response. Magnephy develops validated assays applicable to a spectrum of tissues and diseases. This represents the strongest scenario, where assay validation and long-term customers create the basis for a scalable revenue strategy.

Scenario B: Premium Specialist CRO

Magnephy remains primarily a highly specialized imaging CRO. Customers pay for individual studies, but projects remain relatively specific to customers, meaning that no pipeline applicable to multiple projects without major changes is developed. The business can still be profitable and establish itself as a niche provider within, for example, the KTH-KI campuses. It nevertheless limits the probability of delivering a return to a venture fund.

Scenario C: Technology Commoditization

Tissue clearing, light-sheet imaging, STED, AI segmentation and 3D analysis become increasingly standardized. Large microscopy companies, CROs and pharmaceutical companies internalize the capabilities. Magnephy's technical advantage narrows and customers increasingly purchase equipment/software rather than outsourcing the workflow. In this scenario, the company becomes an expert service provider without a significant technological moat.

5. Moat Analysis

Magnephy's defensibility is best understood as a stack of four layers.

Technology/IP: not very strong

The methodology has substantial scientific novelty and a well-documented publication history. However, individual components such as tissue clearing, STED microscopy and AI segmentation are not protected by a patent. Importantly, the company states that its Vinnova-backed project included development of its IP strategy, but the scope and strength these patents will have is at the moment unknown and thus not yet an established competitive advantage.

Protocol, know/how: strong

The decade-long academic development behind Magnephy provides accumulated knowledge in all the areas that are key for 3D imaging, which requires expertise rather than just access to a STED microscope.

Application-specific assays: potentially very strong

If Magnephy develops validated assays for particular pharmaceutical questions, these could become substantially more defensible than generic imaging.

AI/ data analysis: potentially the strongest in the long-term

Over time, Magnephy could accumulate a portfolio of key information such as annotated images, biological phenotypes, segmentation models, drug-response patterns or assay validation data that could be used as a backbone to build improved models. This advantage is nevertheless still at an early stage.

6. How the thesis could break down

Layer	Primary Failure Risk
Technical	STED resolution does not translate into sufficiently useful information at the tissue depths, volumes or throughput required by pharmaceutical customers as well as not being adaptable to other tissues beyond kidney
Throughput	High-resolution 3D STED is too slow or expensive for routine pharmaceutical studies
AI	Segmentation does not generalize sufficiently across tissues, diseases and sample preparations, requiring extensive manual assay development
Capital	Expansion requires substantial investment in microscopy equipment and specialized personnel before customer utilization is sufficiently high

7. Key Catalysts

Near-term

- Conversion of pilot projects into recurring pharmaceutical contracts
- Transforming pilot studies into R&D contracts for pharmaceutical companies with expansion of client portfolio
- Ability to keep same customers long term
- Expansion of tissue processing beyond kidney
- Demonstration that 3D measurements lead to a significant improvement in the decisions made from this sort of technology by the pharmaceutical companies.
- Successful securing of further funding to ensure commercial expansion

Medium-term

- Long-term agreements with major pharmaceutical companies
- Standardised assays that are offered to multiple customers.
- Evidence that Magnephy's technology contributes positively to drug-testing decisions on whether to continue it
- Expansion from kidney into oncology, immunology, CNS or other tissues

8. Investment Thesis

Magnephy is a high-risk, high-upside bet on nanoscale 3D tissue imaging becoming a commercially important layer of the R&D pipeline. The science behind the start-up is mature and it is built on more than a decade of peer-reviewed research, which shows that the tissue imaging pipeline works, at least in kidney tissue. The key question is whether pharmaceutical company problems such as studying drug localization, target engagement, biodistribution and subcellular treatment response can be resolved with Magnephy's technology.

The current approach of offering a service is interesting because it allows Magnephy to determine which tissues are suitable for its pipeline, as well as seeing which tissues are more worth focusing resources on, based on the interest they generate. If it develops

standardized assays, proprietary datasets, AI models and recurring pharma relationships, the business can become substantially more scalable and defensible. I would currently treat Magnephy as a promising but commercially unproven deep-tech platform, where the next financing should be contingent on evidence of repeatable customer demand rather than further proof that the microscopy works.

9. When to change your mind

The investment case weakens materially if several of these remain true over the next 24–36 months:

- No meaningful growth in recurring pharmaceutical customers
- Revenue remains dominated by one-off pilot projects
- No major pharmaceutical partnership or framework agreement
- Customers use Magnephy primarily for visualization rather than decision-critical measurements
- Expansion beyond kidney fails to produce commercially viable assays
- STED imaging proves too slow or expensive for routine pharmaceutical use
- Competitors provide sufficiently similar biological information using cheaper light-sheet or conventional imaging
- AI segmentation model becomes unreliable and requires manual adjustments
- Gross margins fail to improve as revenue scales
- No meaningful proprietary assay/data/IP business advantage develops

That would indicate that the technology is valuable scientifically but has not yet become economically indispensable.

10. Bull Case

- Magnephy becomes the leading specialist provider of nanoscale 3D tissue imaging for pharmaceutical R&D
- Multiple major pharmaceutical companies adopt the platform
- Initial kidney specialization becomes a beachhead rather than a limitation
- Standardized assays are developed for antibody therapeutics, oncology, immunology and other modalities
- 3D drug-target colocalisation becomes a recognized preclinical development tool
- AI models and proprietary datasets create increasing analytical differentiation
- Recurring customer programs reduce reliance on one-off CRO projects
- Protocol know-how and validated assays create meaningful switching costs
- The company expands from services into a software/data/assay platform

11. Bear Case

- R&D customers do not generate recurring demand
- Other types of microscope technology, like light-sheet microscopy, provide good enough resolution for companies at a fraction of the cost
- Tissue processing protocol becomes too variable between tissues for it to be reliably introduced as part of an R&D pipeline.

- Expansion beyond kidney tissues fails and Magnephy remains a niche specialised start-up.
- R&D companies come up with their internal imaging processing pipeline

12. Conclusion

The key question on whether Magnephy can be a successful investment relies on determining whether nanoscale 3D imaging can become an integrated analytical layer in pharmaceutical development, and whether Magnephy can position itself as a market reference in the field. The scientific foundation is strong, and the founders have well-established expertise, which ensures that the start-up can deliver what it promises, with early commercial validation also pointing in the right direction. The key question is whether the highly specialised service provided by Magnephy is sustainable long-term. One needs to bear in mind that although the current team has substantial experience, as the company grows they will need to ensure that the quality of the analysis remains consistent and translatable to other tissue samples. This needs to be accompanied by recurring orders from pharmaceutical companies that rely on Magnephy to make key decisions regarding their products. Thus, the technology is compelling, but the venture case remains unproven.